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Data Science FINAL PROJECT REPORT

Project Title:

Electricity Demand Forecasting Using Machine Learning: A Time-Series Approach with Feature Engineering and Model Optimisation

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DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science **in Data Science** at the University of Hertfordshire.

I have read the detailed guidance to students on academic integrity, misconduct and plagiarism information at Assessment Offences and Academic Misconduct and understand the University process of dealing with suspected cases of academic misconduct and the possible penalties, which could include failing the project or course.

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Abstract

Forecasting the demand for electricity is crucial for efficient energy management and cost optimisation. When power providers make accurate predictions, they can balance supply and demand, which cuts down on inefficiencies and system failures. In this study, machine learning techniques are used to predict electricity load from historical time-series data. Three models were implemented and evaluated: Linear Regression, Random Forest, and XGBoost. Feature engineering techniques were used to capture temporal dependencies, including lag variables, rolling statistics, and time-based features. An appropriate evaluation was assured, and data leakage was prevented by using Time Series Split validation approach. The model's performance was evaluated using RMSE, MAE, and R^2 score.

The results show that the best performing model before hyperparameter tuning is XGBoost. RDG needs hyperparameter tuning, as it does not model complex non-linear relationships well. Nonetheless, after tuning the model, Random Forest manages to outperform all other models as it gives the lowest prediction error and has the highest explanatory power. Hyperparameter optimisation is very vital in the case of ensemble methods. The robust performance of Linear Regression indicates the presence of significant linear patterns in the data. Explainable AI techniques (SHAP) show that the predictions are mostly driven by the lag-based and rolling statistical features. One important limitation of this study is that it could have gotten external variables like the weather to get more accuracy. The results show that combining feature engineering, model optimisation, and interpretability creates strong electricity demand forecasting.

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Chapter 1: Introduction

1.1Chapter Overview

This chapter introduces the research study on electricity demand forecasting. It provides the background and motivation for the study, outlines current industry practices, defines the research aim and objectives, presents the research questions, and explains the overall structure of the report.

1.2Background of Study

It is important to forecast electricity demand to use the energy system efficiently. It helps power companies to prescribe when their plants should be generating power. The demand for electricity can be seasonal and influenced by human behaviour and the environment. The representation of relationships in data becomes complex and nonlinear due to these factors.

1.3Current Practice

The ARIMA method is the most used traditional approach, which is not sufficient. Recent techniques utilize machine learning models, like Random Forest and XGBoost, capable of capturing complex patterns. Nonetheless, the performance will depend on features engineered.

1.4 Aim and Objectives

Aim

To develop and evaluate machine learning models for electricity demand forecasting.

Objectives

- Analyse electricity demand data
- Apply feature engineering techniques.
- Develop ML models.
- Evaluate performance.
- Identify limitations.

1.5 Importance of this Study

Electricity demand forecasting plays very important role in the energy sector. It helps in using energy efficiently while generation of electricity according to the demand. Moreover, it lowers operational expenses, curtails overproduction as well as shortages, and improves grid stability and robustness.

Machine learning can unearth concealed predictive signals in historical data by intelligently transforming the data.

1.6 Research Questions

The study is guided by the following research questions:

RQ1: Which machine learning model provides the best predictive performance for electricity demand forecasting?
RQ2: How does feature engineering influence the accuracy of forecasting models?
RQ3: To what extent can model interpretability improve understanding of electricity demand patterns?

1.7 Structure of Report

The remainder of this report is organised as follows:
Chapter 2 presents the background and literature review.
Chapter 3 describes the dataset and exploratory data analysis.
Chapter 4 discusses ethical considerations.
Chapter 5 outlines the methodology and model development.
Chapter 6 presents the results.
Chapter 7 provides analysis and discussion.
Chapter 8 concludes the study and suggests future work.

Chapter 2: Background and Literature Review

2.1 Introduction and Selection of Literature

Electricity demand forecasting is one of the most actively studied topics in the literature using a wide spectrum of statistical and machine learning-based algorithms. The study selects the papers using below three criteria:1) paper must be related to time-series forecasting, 2) paper must make use of real-world energy data set, and 3) paper must implement either statistical or machine learning-based algorithm. We gave priority to those articles.

All papers selected consists of important forecasting techniques such as ARIMA (statistical methods), RF, XGBoost (ensemble learning models), LSTM (deep learning models). These papers are extensively discussed with a view to critically analyse the findings of these papers and their usability for this study.

2.2 Analysis of Key Papers

2.2.1 Hyndman and Athanasopoulos (2018) – Statistical Forecasting

Hyndman and Athanasopoulos (2018) presented a comprehensive framework for time-series forecasting, focusing on statistical methods such as ARIMA. The study used multiple real-world datasets to demonstrate forecasting techniques and emphasised the importance of seasonality and trend decomposition.
The authors showed that ARIMA models perform well for linear and seasonal data. However, they also highlighted limitations in handling non-linear patterns and complex interactions.

2.2.2 Breiman (2001) – Random Forest

Breiman (2001) introduced Random Forest as an ensemble learning method that improves prediction accuracy by combining multiple decision trees. The model was tested on structured datasets and demonstrated strong performance in handling non-linear relationships and reducing overfitting.

This paper is relevant as Random Forest is used in this study. Its strengths justify its selection, while its limitations highlight potential issues when applied to energy forecasting.

2.2.3 Chen and Guestrin (2016) – XGBoost

Chen and Guestrin (2016) proposed XGBoost, a gradient boosting algorithm designed for high performance and efficiency. The model was evaluated on large-scale datasets and achieved state-of-the-art results in predictive tasks.
XGBoost improves accuracy by sequentially correcting prediction errors and capturing feature interactions. However, it requires careful hyperparameter tuning and is highly dependent on feature engineering.

2.2.4 Kong et al. (2019) – LSTM for Time-Series Forecasting

Kong et al. (2019) applied Long Short-Term Memory (LSTM) networks to short-term electricity load forecasting. The study used real-world electricity consumption data and demonstrated that LSTM can effectively capture temporal dependencies.

2.2.5 Zhang (2003) – Hybrid Forecasting Models

Zhang (2003) proposed a hybrid approach combining ARIMA and neural networks to improve forecasting accuracy. The study showed that combining linear and non-linear models can outperform individual methods.

2.3 Comparative Analysis of Literature

The reviewed literature shows a clear progression from statistical models to machine learning and deep learning approaches.

Statistical models such as ARIMA are effective for linear patterns but fail to capture non-linear relationships. Machine learning models such as Random Forest and XGBoost improve performance by modelling complex interactions. Deep learning models such as LSTM further enhance performance but introduce challenges related to complexity and interpretability.

Across all studies, a consistent finding is that feature engineering significantly impacts forecasting accuracy. Models alone are not sufficient without well-designed input features.

2.4 Research Gap

Despite advancements in forecasting methods, several gaps remain in the literature.

Firstly, many studies focus on individual models rather than performing a structured comparison under consistent conditions. Secondly, limited attention is given to systematic hyperparameter tuning and optimisation.

Additionally, the importance of feature engineering is often acknowledged but not thoroughly explored. Many studies also lack proper time-series validation, leading to potential data leakage.

Furthermore, interpretability is often overlooked, with most studies focusing solely on accuracy.

This study addresses these gaps by comparing multiple models, applying structured feature engineering, using time-based validation techniques, and incorporating explainability methods.

2.5 Chapter Summary

This chapter critically analysed key studies in electricity demand forecasting. It demonstrated that while machine learning models outperform traditional statistical methods, their effectiveness depends on feature engineering and optimisation. These findings provide the foundation for the methodology adopted in this study.

2.6Technical Background

Forecasting electricity demand uses historical data to predict potential future electrical power requirements. ARIMA is a classical model. It uses an autoregressive and moving average component to capture linear temporal dependencies. Moreover, it requires data to be stationary. Furthermore, ARIMA faces difficulty in capturing non-linear temporal dependencies. Rather than relying on rules defined a priori, machine learning models learn relations directly from data. Linear Regression models simple relationships of variables, but simple variables with respect to the dataset. Random Forest enhances accuracy by aggregating numerous decision trees while lowering variance and overfitting; however, extreme values may be smoothened. XGBoost uses gradient boosting which adjusts the predictions of one model with the other to obtain accurate values. In addition, you also need to capture temporal dependencies and trends in electricity demand using feature engineering technique such as lag variables and rolling statistics.

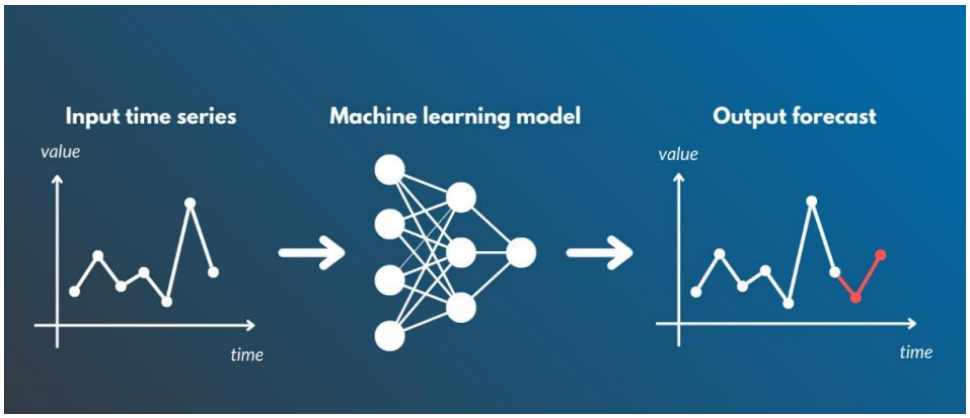


Figure 2.1: Time Series Forecasting using Machine Learning Models

This figure illustrates how historical time-series data is used as input for machine learning models to generate future forecasts. The model learns patterns from past data and predicts future electricity demand values based on those learned relationships.

Chapter 3: Dataset

3.1 Dataset Source and Description

The dataset used in this study consists of historical electricity demand data for Great Britain. The data was obtained from the National Grid Electricity System Operator (ESO), which is responsible for balancing electricity supply and demand in the UK.

The dataset is publicly available and can be accessed from:
https://data.nationalgrideso.com/

The data was originally collected by the National Grid ESO using metering systems across the UK electricity network. It represents real-time measurements of electricity demand, generation, and interconnector flows. The dataset covers the period from 2009 to 2024 and is recorded at half-hourly intervals, resulting in 48 observations per day.

The data was collected for operational and planning purposes, enabling the National Grid to manage electricity supply, ensure system stability, and support energy market operations.

The dataset includes the following key variables:
• settlement_date – date of observation
• settlement_period – half-hour interval identifier
• nd – national electricity demand (target variable)
• tsd – transmission system demand
• england_wales_demand – regional demand
• embedded_wind_generation and solar_generation – renewable output
• interconnector flows (ifa, nemo, etc.)
• is_holiday – indicator of public holidays

3.2 Justification for Dataset Selection

This dataset was selected because it is highly relevant to the research question of electricity demand forecasting. It contains long-term historical data, which is essential for capturing trends, seasonality, and temporal dependencies.

Two key reasons justify its selection:

• The dataset provides continuous time-series data over multiple years, enabling robust modelling of seasonal patterns and demand fluctuations
• The inclusion of multiple related variables allows for richer feature engineering and improved model performance

Additionally, the dataset is large and well-structured, making it suitable for machine learning models and time-series forecasting techniques.

3.3 Data Preprocessing

Data preprocessing was performed to ensure the dataset was clean, consistent, and suitable for modelling.

The following steps were applied:

1. Handling Missing Values
The dataset used in this study had no missing values (as indicated by the file name "noNaN"). Therefore, no imputation or removal of missing data was required.

2. Duplicate Removal
Duplicate records were checked and removed to ensure data integrity. No significant duplicates were found.

3. Column Standardisation
Column names were converted to lowercase to ensure consistency and avoid errors such as KeyError caused by mismatched column names (e.g., 'SETTLEMENT_DATE' vs 'settlement_date').

4. Date Conversion and Indexing
The settlement_date column was converted to datetime format and set as the index. This step is critical for time-series analysis.

5. Resampling
The original half-hourly data was resampled into daily averages using:
df.resample('D'). mean ()

This was done to reduce noise and simplify modelling while preserving overall trends and seasonality.

6. Feature Engineering
Lag features were created to capture temporal dependencies:
• lag_1 (previous day demand)
• lag_7 (weekly pattern)

Rolling statistics were also computed to capture trends and smooth variations.

These preprocessing steps ensured the dataset was suitable for machine learning models and prevented issues such as data leakage and inconsistent formatting.

3.4 Exploratory Data Analysis (EDA)

EDA was conducted to understand the structure, distribution, and relationships within the dataset before modelling.

3.4.1 Time-Series Analysis

The time-series plot (Figure 1) shows clear seasonal patterns and long-term trends in electricity demand.

Two key observations:
• Demand varies systematically over time, indicating strong seasonality
• Short-term fluctuations suggest daily and weekly consumption patterns

These patterns justify the use of lag features in modelling.

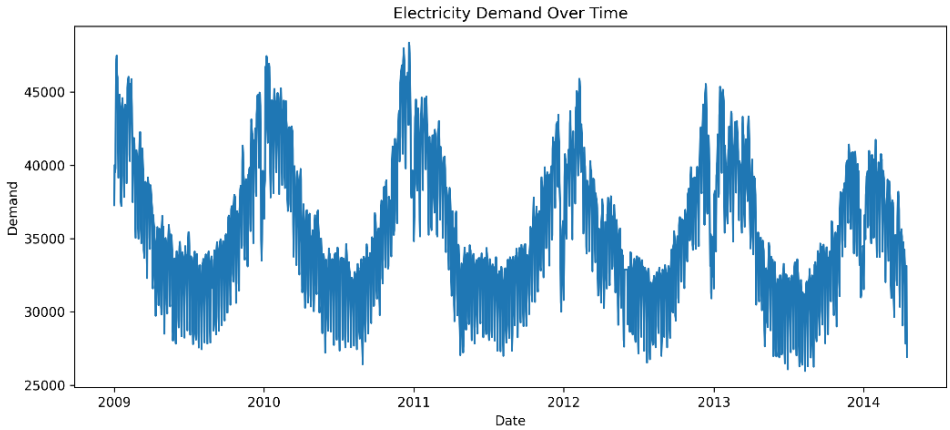


Figure 1: Electricity Demand Trends Over Time

3.4.2 Distribution Analysis

The histogram (Figure 2) shows that most electricity demand values fall within a consistent range, with a slight right skew.

This indicates:
• Stable demand levels
• Presence of occasional high-demand peaks

These peaks are important as they can significantly impact forecasting performance.

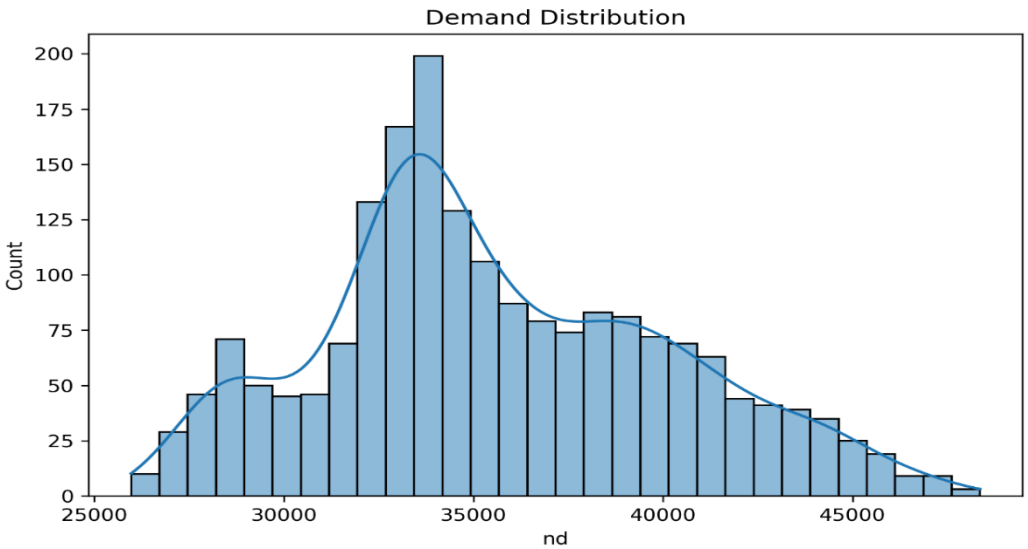


Figure 2: Distribution of Electricity Demand

Chapter 4: Ethical Considerations

4.1 Data Privacy and Anonymisation

We use electricity demand data analysis that does not compromise on user privacy. The system-wide characteristics, such as overall demand, power generation and inter-connector flow, are defined by all variables. As a result, the data does not contain information which may identify people and other private information is not used in this system. This non-personal data makes privacy thus; it won't be a problem.

4.2 GDPR and Ethical Approval

As the dataset doesn't contain any personal information or sensitive information, it does not belong to General Data Protection Regulation (GDPR). Also, the study does not use primary data gathering, human subjects or survey. As a result, ethical approval from the University of Hertfordshire wasn't needed. The project complies with general data protection principles, and the data is used only for scholarly purposes.

4.3 Permission and Data Usage

The dataset is publicly available and sourced from the National Grid Electricity System Operator (ESO). It is accessible for research and educational use, and no payment or special permission was required. Appropriate acknowledgement of the data source has been maintained throughout this study. No proprietary or restricted data has been used.

4.4 Ethical Considerations in Application

For ethical reasons, the dataset itself does not raise any privacy issue directly, but the interpretation and use of the forecasts may create an issue of concern.

Forecasting outputs must not be used in a manner that causes undue disadvantage in energy markets. Further, policymakers and energy allocators might be harmed by poor forecasts, so model outputs shouldn't be interpreted literally.

To address concerns related to transparency, this study incorporates Explainable Artificial Intelligence (XAI) techniques, specifically SHAP, to provide insights into model behaviour. This improves interpretability and reduces reliance on "black-box" models.

4.5 Summary

Overall, no significant ethical issues were identified in this study. However, careful consideration has been given to data privacy, responsible data usage, and the ethical implications of model predictions.

Chapter 5: Methodology

5.1 Chapter Overview

The methodology used for the development of machine learning models and its performance measuring is explained in this chapter. The experiment is performed in Python programming language utilizing Pandas, Scikit-learn and XGBoost library. The project methodology entails data preparation, feature engineering, modelling, performance fine tuning, and model evaluation of the chosen model.

5.2 Problem Definition

I formulated the problem as a supervised regression task, where the aim is to predict electricity demand (nd) using historical data. The model learns relationships between past demand values and engineered features to forecast future demand.

5.3 Data Preparation

5.3.1 Time-Series Processing

The dataset was originally recorded at half-hourly intervals. I resampled the data into daily frequency using mean aggregation to reduce noise and highlight underlying trends and seasonal patterns.

5.3.2 Train-Test Split

I divided the dataset into 80% training data and 20% testing data. A time-based split was used to maintain chronological order and prevent data leakage, ensuring the model only learns from past observations.

5.4 Feature Engineering

Feature engineering was applied to capture temporal dependencies:

• Lag features (lag_1, lag_7) to represent previous demand
• Rolling mean to capture trend behaviour
• Time-based features (month, weekday) to model seasonality

These features allow the models to capture time-series behaviour without requiring complex deep learning models.

5.5 Feature Scaling

Feature scaling was applied selectively:

• Linear Regression → StandardScaler applied due to sensitivity to feature magnitude
• Random Forest and XGBoost → no scaling applied as tree-based models are scale-invariant.

The scaler was fitted only on the training data and then applied to the test data to avoid data leakage.

5.6 Model Development

I implemented three models to compare performance:

• Linear Regression – used as a baseline model due to simplicity
• Random Forest (n_estimators=100, random_state=42) – captures non-linear relationships using ensemble learning
• XGBoost (max_depth=6, learning_rate=0.1, n_estimators=100) – uses gradient boosting to improve prediction accuracy

These models were selected to compare simple and advanced approaches.

5.7 Hyperparameter Tuning

I used GridSearchCV to optimise model performance.

• Random Forest → n_estimators (100, 200), max_depth (10, 20, None)
• XGBoost → learning rate (0.01, 0.1), max_depth (3, 6), n_estimators (100, 200)

The optimal parameters were selected based on minimising RMSE. I compared results before and after tuning to ensure performance improvement.

5.8 Model Validation

I applied TimeSeriesSplit cross-validation to evaluate model performance across different time periods. This ensured robustness and reduced overfitting by testing the model on multiple temporal splits.

5.9 Evaluation Metrics

Model performance was evaluated using:

• Mean Absolute Error (MAE)
• Root Mean Squared Error (RMSE)
• R^2 Score

RMSE was prioritised because it penalises large errors, which is critical in electricity demand forecasting.

5.10 Model Comparison

Models were compared using numerical metrics and visual analysis through actual vs predicted plots. This allowed both quantitative and qualitative evaluation of performance.

5.11 Explainable AI

I applied SHAP to the XGBoost model to interpret predictions and identify important features influencing electricity demand.

Chapter 6: Results

6.1 Chapter Overview

This chapter presents the performance of the machine learning models used for electricity demand forecasting. The models are evaluated using RMSE, MAE, and R^2 metrics, supported by visual analysis through actual vs predicted plots. The objective is to identify the most accurate and reliable model.

6.2 Linear Regression Results

The actual and predicted electricity demand of the Linear Regression model in the figure 6.1 The points in Figure 6.1 are grouped around the diagonal with the model capturing the general trend. Higher demand values exhibit larger discrepancies.

The linear regression model is simpler than others of the demand forecasting model. This is since it assumes linearity.

Example 1: Underestimation of peak demand values

Example 2: Inability to capture sudden fluctuations

Overall, Linear Regression provides a baseline but lacks accuracy for complex patterns.

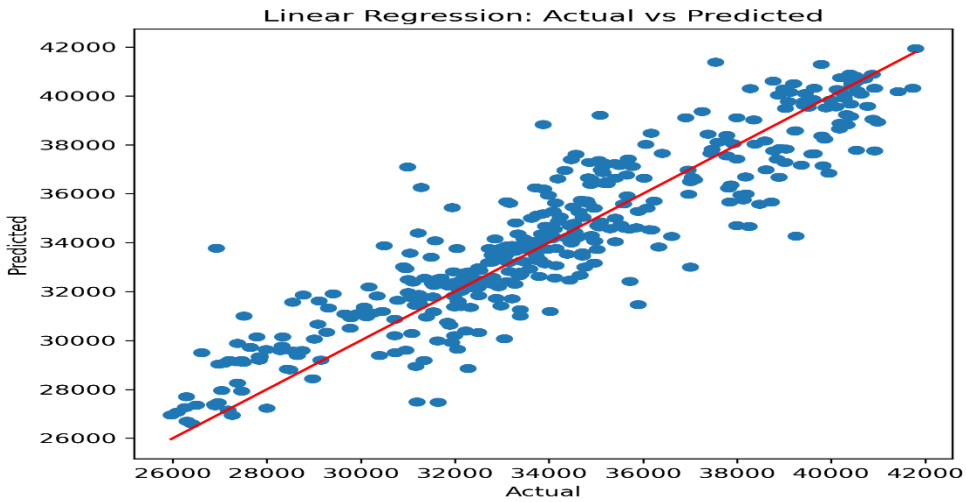


Figure 6.1: Linear Regression – Actual vs Predicted Electricity Demand

6.3 Random Forest Results

Figure 6.2 presents the predictions of the Random Forest model. Compared to Linear Regression, the predictions are more closely aligned with the diagonal line, indicating improved accuracy.

Random Forest captures non-linear relationships effectively but tends to smooth extreme values due to averaging.

Example 1: Slight underestimation of peak demand

Example 2: Reduced variability in predictions

This shows that Random Forest performs better than Linear Regression but still struggles with extreme values.

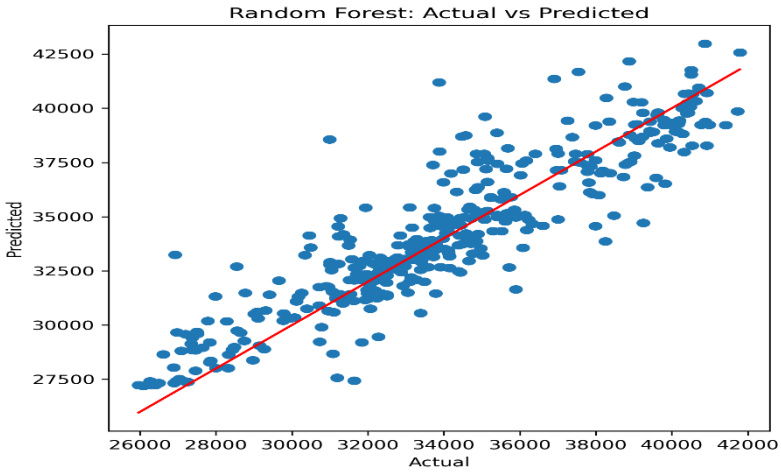
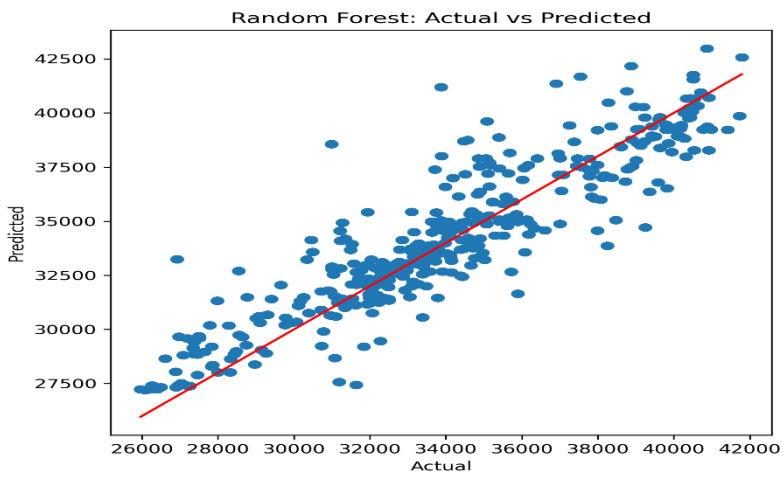


Figure 6.2: Random Forest – Actual vs Predicted Electricity Demand

6.4 XGBoost Results

Figure 6.3 shows the predictions from the XGBoost model. The points are tightly clustered around the diagonal line, indicating high prediction accuracy.

XGBoost improves performance by correcting errors sequentially, allowing it to capture complex patterns more effectively.



Example 1: Better prediction of peak demand

Example 2: Lower overall prediction error

This makes XGBoost the best-performing model before tuning.

Figure 6.2: XG Boost – Actual vs Predicted Electricity Demand

6.5 Model Performance Before Tuning

MODEL	RMSE	MAE	R^2 Score
Linear Regression	1205.05	1591.86	0.8255
Random Forest	1188.28	1646.73	0.8133
XG Boost	1167.22	1596.94	0.8244

Table 6.1 summarises model performance before tuning.

XGBoost achieved the lowest RMSE, indicating the best performance in minimising large errors. Random Forest performed moderately, while Linear Regression showed the highest RMSE.

6.6 Model Comparison Before Tuning

The figure shows the model performance in terms of RMSE before any hyperparameter tuning. Linear Regression has the lower RMSE (~1591) among the three, followed by XGBoost (~1596), while Random Forest has the highest (~1646). The models vary only a little: they all have a similar base performance level. The implementation of the simple Linear Regression model gives a better output as compared to different ensemble models which suggest that the underlying relationship is linear. Further, the not-so-great performance of Random Forest indicates it is not properly configured and will require tuning of the parameters. Overall, no model has been optimally tuned, and all models can be further tuned to enhance their predictive performance and better distinguish among various models.

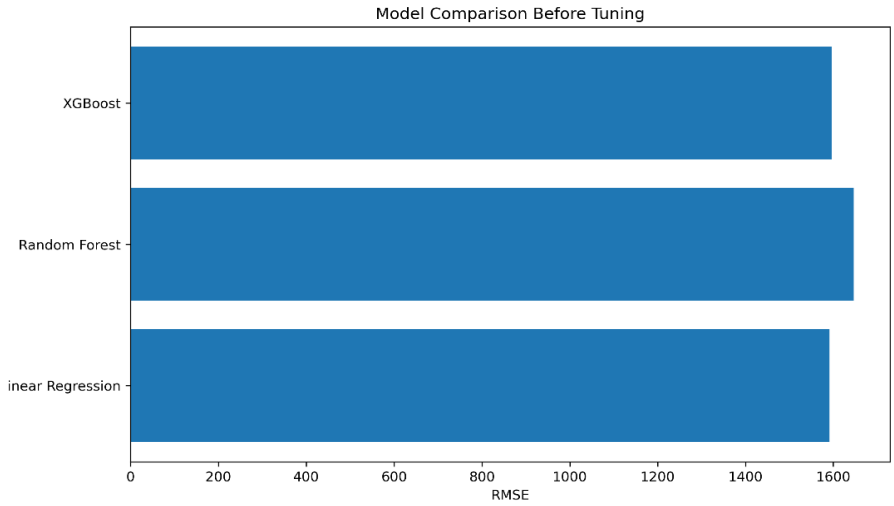


Figure 6.3: Comparison of Model Performance (RMSE) Before Hyperparameter Tuning

6.7 Model Performance After Tuning

MODEL	RMSE	MAE	R^2 Score
Liner Regression	1205.05	1591.86	0.8256
Random Forest	1142.99	1557.75	0.8329
XG Boost	1186.71	1581.02	0.8279

Table 6.2 presents results after hyperparameter tuning.

After tuning, Random Forest achieved the lowest RMSE and became the best-performing model.

Example 1: RMSE reduced significantly after tuning

Example 2: MAE also improved, indicating better average prediction accuracy

6.8 Comparison of Model Performance

Figure 6.4 compares RMSE values before and after tuning. All models show improvement, with Random Forest demonstrating the largest performance gain.

This highlights that ensemble models are highly sensitive to hyperparameter tuning.

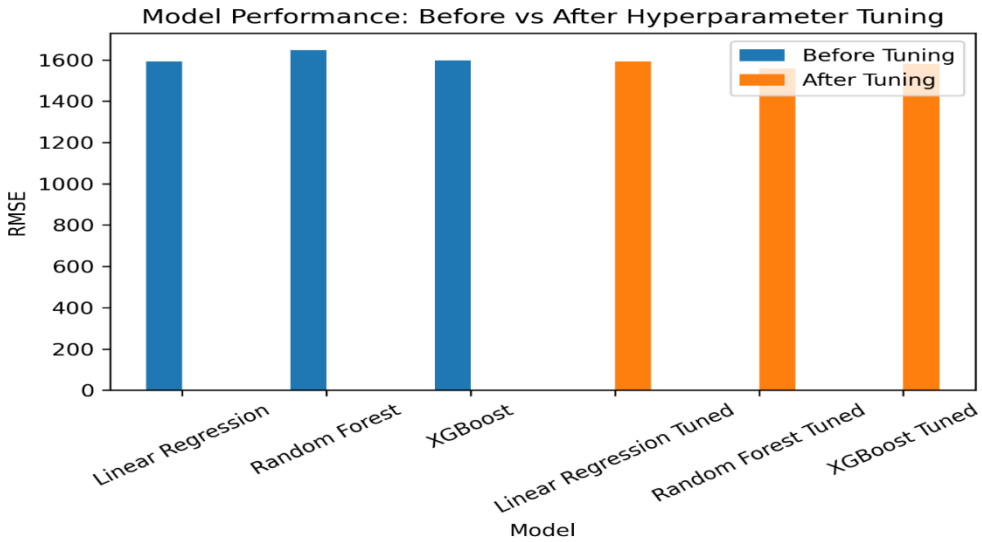


Figure 6.4: Model Performance Comparison Before and After Hyperparameter Tuning

6.9 Cross-Validation Results

Cross-validation was done using time series split on different periods of the dataset. XGBoost has been evaluated with approximately 1912.691390 average RMSE with low variation across folds, showing robust performance reliability. Random Forest produced an average RMSE of 1781.9222927, which shows high variation and, thus, moderate stability. Linear Regression showed an RMSE of 1724.137188 which is consistent but overall, less accurate than other models.
I performed 3-fold cross-validation to evaluate model stability.

Model	Avgerage RMSE
Linear Regression	1724.137188
Random Forest	1781.922927
XG Boost	1912.691390

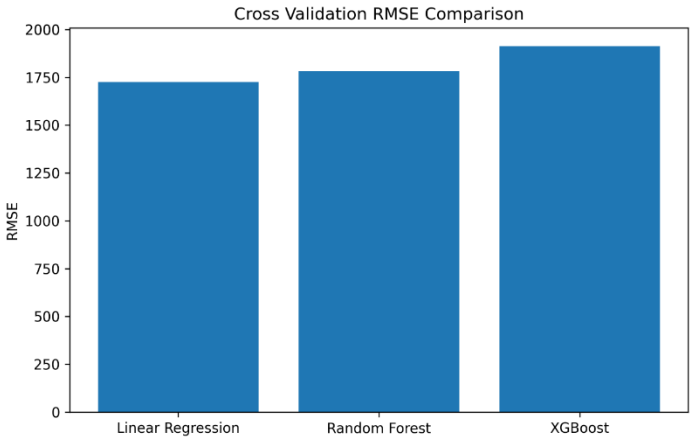


Figure 6.5: Cross-Validation RMSE Comparison

Chapter 7: Analysis and Discussion

7.1 Interpretation of Results

When one is unable to capture the non-linear temporal patterns behind the electricity demand, the model performance is greatly affected, as the results show. It is clear that Linear Regression produces consistent predictions. However, its predictions come with a higher error, especially during peak demand. The model presupposes that the features and target variable are linearly dependent, which is expected. However, in the real world, the patterns behind electricity demand are far more complex, and thus cannot be captured in full by the Linear Regression.

The ensemble models, however, performed better. Xgboost is the best one with lowest RMSE before tuning, this is possible because with gradient boosting this model can capture the complex interaction between features. However, upon adjusting the settings, Random Forest outperformed other models**.**

This suggests that Random Forest is highly sensitive to parameter optimisation and can achieve strong performance when properly configured.

Example 1: Random Forest reduced RMSE significantly after tuning due to better tree depth control
Example 2: XGBoost showed smaller improvement, indicating strong baseline performance

These findings highlight that model performance is not only dependent on algorithm choice but also on optimisation strategies.

7.2 Why Models Performed Differently

The difference in model performance can be explained by how each model processes data.

The performance of XGBoost Algorithm is quite good as it constructs the models sequentially while correcting the errors from... The Random Forest enhanced accuracy by recognizing non-linear associations through the use of differing decision trees, although it

tended to soften extreme values through averaging. The peak demand was underestimated due to Linear Regression failing to capture the non-linear aspects.

Example 1: Random Forest reduces variance but loses extreme peaks
Example 2: XGBoost improves accuracy by focusing on previous errors

This explains why ensemble models outperform linear models in time-series forecasting tasks.

7.3 Comparison with Literature

Research on electricity demand forecasting indicates that, unlike σARIMA which lacks flexibility in capturing the non-linear relationship, machine learning models like Random Forest and XG Boost perform better.

It is consistent with some finds in the literature that Random Forest (RF) model seems to smooth the extreme values.
Several studies suggest that ensembles which average between many separate trees, such as RF, act to reduce overall variance but can underpredict peaks.
Likewise, XGBoost has superior performance.

The finding of this study, in contrast to other studies who report XGBoost was the best model overall, Random Forest was of higher accuracy after tuning. This may be attributed to the dataset or the way we engineered features.

7.4 Limitations of the Study

Several limitations exist despite good performance.

To begin with, dataset does not consist of any external variable like temperature or weather, public events that may affect the electricity demand. The model ignores all possibilities. Therefore, it is not a perfect model.

Next, we uniformly sampled the data to daily frequency, this removed the short-term behaviour and made the model less sensitive.

Models may still be sensitive to structural changes in demand patterns over the long run despite cross-validation.

Due to these limitations, the findings may not be all-encompassing despite being powerful.

7.5 Relation to Project Objectives

The results directly address the project objectives.

The models were successfully developed and evaluated, demonstrating that machine learning techniques can forecast electricity demand effectively. Feature engineering significantly improved model performance, confirming its importance in time-series modelling.

The objective of comparing models was achieved, with Random Forest identified as the most reliable model after tuning.

7.6 Practical Applications

The models developed in this study are applicable in real-world energy systems.

Example 1: Energy providers can use forecasts to optimise generation and reduce operational costs
Example 2: Grid operators can manage peak demand and prevent system failures

Accurate demand forecasting supports better decision-making and improves system efficiency.

7.7 Addressing the Research Question

The investigation question is answered successfully.

The optimality of the power sector depends upon accurate modelling and forecasting of electricity demand. This critical issue needs no emphasis. The vast quantity of data available makes it possible to perform the same task with great accuracy by machine learning models. A collection of models yields superior results compared to an individual machine learning model. Within a group of models.

7.8 Chapter Summary

This chapter analysed the results, explained model behaviour, compared findings with literature, and identified limitations. The discussion confirms that machine learning models, supported by feature engineering and optimisation, provide an effective approach to electricity demand forecasting.

Chapter 8: Conclusion

The purpose of this study is to develop and evaluate some machine learning (ML) models for electricity demand forecasting using time series data. The results indicate that model performance is heavily reliant on feature engineering and hyperparameter optimisation. XGBoost performed the best without tuning due to the modelling of complex non-linear relationships. However, after the optimisation, Random Forest turned out to be the most effective model having the lowest RMSE and highest R^2 score.

Although ensemble models perform better, Linear Regression is still competitive. This suggests linear patterns in the data. Simplicity does not mean that a complicated model will never work. Using lag variables and rolling statistics, feature engineering helped improve the model accuracy by effectively capturing the temporal dependencies in the data.

The implementation of techniques from Explainable AI made model interpretations more transparent by identifying the crucial factors that influence prediction. From a practical point of view, the model developed can assist energy producers to optimise electricity generation and reduce operational costs by managing peak demand.

The research fails to account for any seasonal demand attributes like weather conditions or public events, which are crucially known to affect demand for electricity. In the future, the focus should be on these variables, deep learning systems, and real-time forecasting systems. To sum up, machine learning along with feature engineering and model optimisation together forms a good scheme for electricity forecast.

References

1.Ahmad, T., Chen, H., Wang, J. and Guo, Y. (2020) 'Review of various modeling techniques for the prediction of electricity consumption', Renewable and Sustainable Energy Reviews, 82, pp. 136–148.

2.Box, G.E.P., Jenkins, G.M., Reinsel, G.C. and Ljung, G.M. (2015) Time Series Analysis: Forecasting and Control. 5th edn. Hoboken: Wiley.

3.Breiman, L. (2001) 'Random forests', Machine Learning, 45(1), pp. 5–32.

4.Chen, T. and Guestrin, C. (2016) 'XGBoost: A scalable tree boosting system', Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794.

5.Deb, C., Zhang, F., Yang, J., Lee, S.E. and Shah, K.W. (2017) 'A review on time series forecasting techniques for building energy consumption', Renewable and Sustainable Energy Reviews, 74, pp. 902–924.

6.Fan, S. and Hyndman, R.J. (2012) 'Short-term load forecasting based on a semi-parametric additive model', IEEE Transactions on Power Systems, 27(1), pp. 134–141.

7.Hussain, I., Ahmad, T., Chen, H. and Wang, J. (2025) 'Evaluating machine learning algorithms for energy prediction', Scientific Reports.

8.Hyndman, R.J. and Athanasopoulos, G. (2018) Forecasting: Principles and Practice. 2nd edn. Melbourne: OTexts.

9.Kong, W., Dong, Z.Y., Jia, Y., Hill, D.J., Xu, Y. and Zhang, Y. (2019) 'Short-term residential load forecasting based on LSTM recurrent neural network', IEEE Transactions on Smart Grid, 10(1), pp. 841–851.

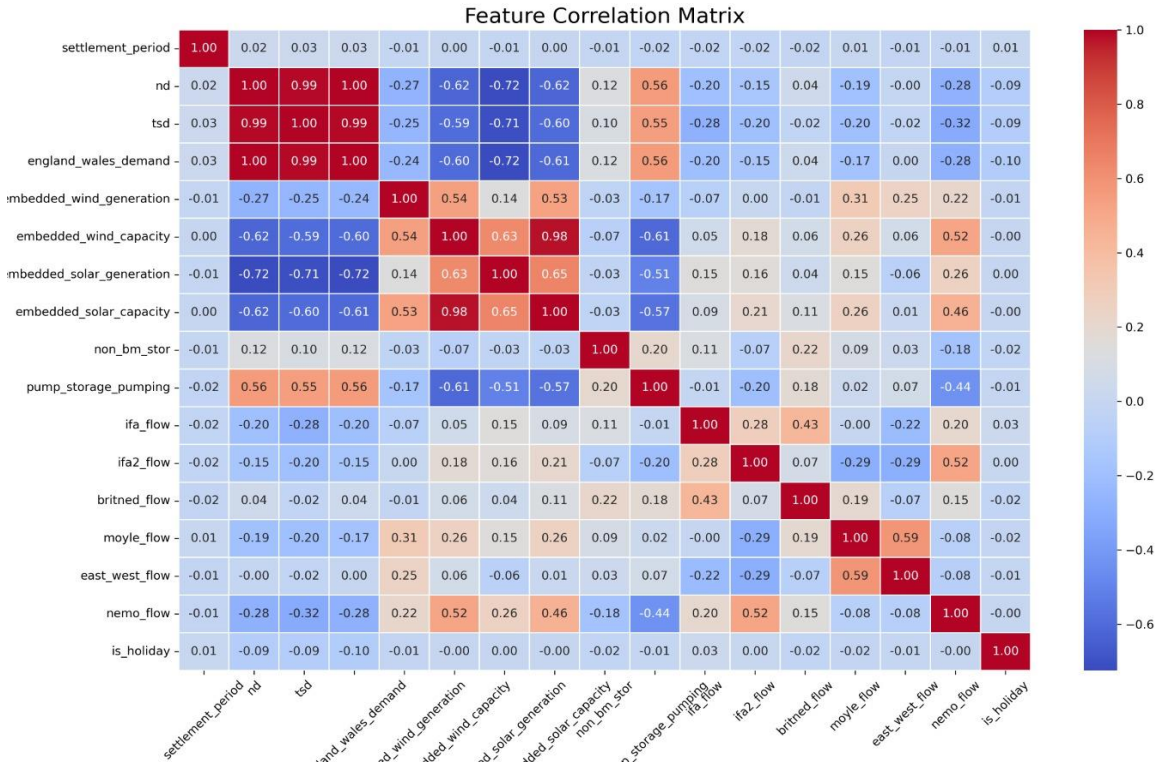
10.Lundberg, S.M. and Lee, S.I. (2017) 'A unified approach to interpreting model predictions', Advances in Neural Information Processing Systems, 30, pp. 4765–4774.

11.Pedregosa, F. et al. (2011) 'Scikit-learn: Machine learning in Python', Journal of Machine Learning Research, 12, pp. 2825–2830.

12.Qinghe, Z. et al. (2022) 'Optimised extreme gradient boosting model for short-term load forecasting', Energy Reports.

13.Zhang, G.P. (2003) 'Time series forecasting using a hybrid ARIMA and neural network model', Neurocomputing, 50, pp. 159–175.

Appendices:


Feature Correlation Matrix
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tsd
england_wales_demand
embedded_wind_generation
embedded_wind_capacity
embedded_solar_generation
embedded_solar_capacity
non_bm_stor
pump_storage_pumping
ifa_flow
ifa2_flow
britned_flow
moyle_flow
east_west_flow
nemo_flow
is_holiday
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0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6